

PSLE MATHEMATICS PRACTICE PAPER 6

Data Handling & Probability

Duration: 1 hour

Total Marks: 50

Name: _____

Date: _____

SECTION A: Short Answer Questions (20 marks)

1. Find the mean of: 5, 8, 10, 7 (2 marks)

2. Find the median of: 3, 7, 5, 9, 1 (2 marks)

3. Find the mode of: 2, 5, 2, 8, 2, 5 (2 marks)

4. What is the range of: 15, 8, 22, 10 (2 marks)

5. How many ways can you arrange 3 different books on a shelf? (2 marks)

6. A spinner has 8 equal sections. What is the probability of landing on one specific section? (2 marks)

7. If a die is rolled, what is the probability of getting an odd number? (2 marks)

8. How many outcomes are there when tossing two coins? (2 marks)

SECTION B: Long Answer Questions (30 marks)

9. The heights of 5 students are 150, 155, 152, 160, 158 cm. Find mean, median and mode. (4 marks)

10. A survey shows: 12 students like maths, 8 like English, 15 like science. Draw a bar graph to represent this data. (4 marks)

11. A bag has 5 red, 3 blue, and 2 green balls. Find the probability of drawing a blue ball. (3 marks)

12. Complete the frequency table and find the mean age. Ages: 10, 11, 11, 12, 10, 10, 13.
(4 marks)

13. If the average of four numbers is 25, and three of the numbers are 12, 22, 28, find the fourth number. (3 marks)

14. A pie chart shows: 40% like pizza, 30% like burgers, rest like salads. Find the percentage who like salads and draw the pie chart. (4 marks)

15. In a class of 30 students, 18 passed maths. Find the percentage who passed and who failed. (3 marks)

GCSEMathsAI

SOLUTIONS

SECTION A SOLUTIONS

Q1: Mean of 5, 8, 10, 7

Mean = $(5+8+10+7) \div 4 = 30 \div 4 = 7.5$

Q2: Median of 3, 7, 5, 9, 1

Arrange: 1, 3, 5, 7, 9. Median = 5

Q3: Mode of 2, 5, 2, 8, 2, 5

2 appears 3 times (most frequent), Mode = 2

Q4: Range of 15, 8, 22, 10

Range = $22 - 8 = 14$

Q5: Arrange 3 books

$3! = 3 \times 2 \times 1 = 6$ ways

Q6: Spinner probability

$P = 1/8$

Q7: Odd number on die

Odd numbers: 1, 3, 5. $P = 3/6 = 1/2$

Q8: Two coins outcomes

HH, HT, TH, TT = 4 outcomes

SECTION B SOLUTIONS

Q9: Mean, median, mode

Mean = $(150+155+152+160+158) \div 5 = 155$, Median = 155, Mode = None

Q10: Bar graph

Draw bars for 12, 8, 15 for maths, English, science

Q11: Probability of blue

Total balls = $5+3+2 = 10$, $P(\text{blue}) = 3/10$

Q12: Mean age

Ages: 10(3), 11(2), 12(1), 13(1). Mean = $(30+22+12+13) \div 7 = 77 \div 7 = 11$

Q13: Fourth number

Sum of 4 numbers = $25 \times 4 = 100$, Fourth = $100 - 20 - 22 - 28 = 30$

Q14: Pie chart

Salads = $100\% - 40\% - 30\% = 30\%$

Q15: Pass percentage

Passed = $18/30 \times 100\% = 60\%$, Failed = $12/30 \times 100\% = 40\%$

GCSEMathsAI